TRANSLATION OF INDIC LANGUAGES USING TRANSFORMER MODEL

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Abstract- **Translation has long been a valuable tool for communicating across cultures and languages. The requirement for precise and effective translation has become even more crucial as a result of globalisation and increased international interactions. Considering its diverse population and multilingualism, India's translation of Indian languages into English has risen in importance. Indic languages such as Hindi, Kannada, Malayalam, Tamil, Telugu, and others require a thorough study of both the source and target languages, as well as their grammatical structures, nuances, and cultural contexts. Advances in technology and computational linguistics have made substantial progress in the field of machine translation in recent years, allowing for the translation of massive quantities of text with tolerable accuracy.**

Despite these advancements, human translation remains the gold standard, especially for complicated or sensitive content. Skilled translators play an important role in ensuring that the translated content accurately represents the original language's intended meaning, tone, and context.

The goal of this research project is to look at the existing state of translation from Indian languages to English utilising machine translation technologies.

I. INTRODUCTION

In terms of language, India is among the most diverse nations in the world, therefore it can be said that India has been referred to as the "Land of Many Tongues and a tower of veritable languages." India's states had been reorganised linguistically in 1950. The Indian Constitution recognises 22 languages. In total, there are 121 official languages and 270 indigenous languages. In India, 96.71% speak one of the scheduled languages as their first language, whereas 3.29% speak another. This creates a schism between English and India's national languages.

By allowing computers to read, analyse, and synthesise natural language writing, NLP (Natural Language Processing) can be used to improve the translation of Indian languages to English. Machine learning, deep learning, and neural networks have all been successfully used to address language translation challenges, allowing for the construction of accurate and efficient translation models. These models can be trained on massive datasets of Indic language texts in order to learn the language's underlying patterns and structures and enhance translation accuracy. Furthermore, NLP may be utilised to overcome the difficulties provided by the various vocabulary and grammar structures of Indic languages, making it a viable tool for enhancing Indic language translation to English.

Translation is a critical tool for communicating effectively across cultures and languages. With the increasing demand for accurate and efficient translation, researchers have investigated the use of advanced machine learning algorithms such as LSTM (Long Short-Term Memory), BPE (Byte Pair Encoding), and Fairseq for Indic language translation to English.

Hindi, Kannada, Tamil, Telugu, and other Indian languages have complicated grammar structures, rich vocabulary, and distinct cultural settings, making proper translation to English difficult. With its capacity to learn the fundamental patterns of the language and enhance translation accuracy, LSTM, a sort of neural network, has shown significant promise in this respect.

BPE, on the other hand, is a subword tokenization strategy that divides words into smaller pieces to aid in translation. This strategy has been demonstrated to be especially helpful in dealing with out-of-vocabulary words, which are a prevalent problem in Indian language translation.

Lastly, Fairseq is a cutting-edge neural machine translation system that improves translation accuracy via the use of convolutional neural networks and attention approaches. This method outperforms existing machine translation systems, making it a viable tool for Indian language translation.

II. RELATED WORKS

The paper [1] covers the creation of a hybrid methodology for machine translation of the Malayalam language, which employs both rule-based and statistical methods. Preprocessing, morphological analysis, rule-based translation, and statistical post-processing are the four major phases of the suggested approach. The system was tested and compared to existing machine translation systems using the BLEU (bilingual evaluation

understudy) score. The results demonstrated that the suggested hybrid technique outperformed the previous systems, achieving an improvement in Malayalam to English translation accuracy. The work demonstrates the efficacy of integrating rule-based and statistical methods in machine translation and provides useful insights for designing better translation systems for additional languages.

The research in paper [2] assesses the performance of existing Arabic to English machine translation systems using the BLEU metric, which compares machine translation output against a reference translation to determine quality. The research examines three machine translation systems: Google Translate, Bing Translator, and Moses, a statistical machine translation system. In terms of translation accuracy, the Moses system surpassed the other systems, getting the highest BLEU scores. The study sheds light on the strengths and drawbacks of existing Arabic to English machine translation systems, emphasising the importance of ongoing research to improve their accuracy and efficacy.

The study in paper [3] provides a new model for Finnish-Swedish translation using BERT, a pre-trained language model for natural language processing tasks. The proposed model in [3] combines BERT with a sequence-to-sequence neural machine translation architecture trained on a parallel corpus of Finnish and Swedish literature. The model's performance is assessed using the BLEU metric, which measures the quality of the translation output. The BERT-fused model outperformed existing machine translation systems for Finnish-Swedish translation, earning higher BLEU scores. The work [3] illustrates the efficacy of combining BERT with neural machine translation for enhancing language translation accuracy and provides useful insights for constructing superior translation models for other language pairs.The paper [4] emphasises the use of preprocessors and Machine translation from English to Malayalam which is used in Natural Language Processing (NLP) applications. To improve automated translation accuracy, the authors propose a preprocessing pipeline that includes processes such as tokenization, part-of-speech tagging, and stemming. The BLEU measure is used to evaluate the achievement of the proposed preprocessing pipeline.

An overview of the many techniques and methodology used in sentiment analysis and machine translation, covering traditional methods like rule-based approaches and contemporary methods like deep learning techniques was described in paper [5].

The topic of negative expression translation in Japanese and Chinese machine translation is addressed in paper [6]. The authors suggest a strategy for dealing with negative terms in the translation process, such as negation. The features of negative expressions were examined in both languages and proposed a rule-based technique for dealing with them throughout the translation process. The report describes the experiments that were carried out to evaluate the suggested strategy and compares the results to those of other existing methodologies.

The paper [7] describes a novel technique for script translation based on natural language processing (NLP). To translate text from one script to another, the technique employs NLP tools such as sentence tokenization, word sense disambiguation, and part-of-speech tagging. The method is used to convert Marathi text to Hindi script.

The study discovered that the Transformer model beats the other approaches in terms of translation quality, particularly when dealing with extended phrases and uncommon vocabulary. The study also reveals that using pre-trained models and substantial training datasets can improve the Transformer model's translation quality .Transformer machine translation is a promising approach for Thai-English and English-Thai translation and has the potential to be widely used in a variety of language-related applications[8]. The SMT method employed a phrase-based model, whereas the NMT method employed a transformer model. To increase the quality of the translations, the authors also experimented with several preprocessing techniques such as tokenization and stemming.In terms of translation quality, the study [9] discovered that the NMT approach beat the SMT approach, particularly for longer sentences. Preprocessing procedures such as stemming and stop-word removal were also found to increase translation quality for both methodologies [9].

It developed a novel framework for neural machine translation (NMT) of Indian-English languages. The authors propose a two-step process in which the source language is first normalised before being translated into the target language.An Indian language transliteration model is used to translate the source language text into a phonetic representation, which is then mapped to the appropriate English letters using a grapheme-to-phoneme model. The purpose of this strategy is to address the difficulties posed by the intricate writing systems of Indian languages. The created model was trained on a massive parallel corpus of Indian and English literature [10]. .

The strategy is divided into two stages. In the first stage, the authors employ an unsupervised technique to extract a huge number of translation rules. These rules are then filtered based on their frequency and correctness to provide a smaller collection of high-quality rules. In the second stage, the authors employ a supervised learning technique to further optimise the chosen rules. A machine learning model is trained to estimate the quality of each translation rule based on factors such as the frequency of the rule, its length, and the closeness of the source and destination language phrases. The results of their methodology on a Chinese-English translation challenge reveal that it beats numerous baseline approaches in terms of translation quality [11].

One downside of utilising a hybrid-strategy technique that combines semantic analysis with rule-based MT is that it can be computationally costly and may necessitate substantial domain expertise. To handle the particular domain of patent documents, the semantic analyzer and rule-based MT system must be properly built and trained [12].

Tokenizing the incoming English text and transforming it into a series of words completes the pre-processing stage. The SMT module is used to translate the English text into the Indian language of the target. The SMT module employs a phrase-based translation approach. The Moses toolkit, a popular open-source tool for constructing SMT systems, is used to train the model. The NMT module is used as a post-processing step to adjust the final product of the SMT module in order to increase translation quality. Using the Transformer architecture, the NMT module is trained on a huge parallel corpus of English and Indian language utterances. The authors built the NMT module with the OpenNMT toolbox [13].

A new dataset for English to PSL translation, consisting of 1,500 phrases and their PSL translations, may be employed. The model is trained on this dataset using several machine learning methods

such as decision trees, Naive Bayes, and SVMs, and then the source language module preprocesses the input English text using various NLP approaches such as part-of-speech tagging, stemming, and stop-word removal [14]. As the need for NLP grows, so do the applications of NLP in areas such as healthcare, banking, and e-commerce [15] .

III. PROPOSED METHOD

In NLP, the transformer is an architecture that focuses on addressing sequence-to-sequence tasks while easily handling long-term dependencies. Instead of sequence-aligned RNNs or convolution, it uses self-attention to compute representations of its input and output.

The challenge of how to parallelize sequential data emerges. In 2017, the transformer neural network was launched. The network utilises an encoder and decoder design similar to that of RNNs. Input can be fed in parallel is the difference. .For example, in RNN, the inputs are fed one word of the sentence at a time and gives a vector or for the same. This can be inefficient when it comes to a very large dataset. In contrast, there is no idea of time-step for the input in a transformer encoder model, which may pass all the words in a sentence concurrently and calculate the word embeddings simultaneously, which works better for a big dataset. In terms of input embeddings, the goal is to map every word in space into physically closer words. For example, words like “happy, sad, joyful “ are all emotions which could be placed closer in a space, and words like dog, cat, deer can be placed closer.This can be done either by pretraining an embedding space or using a pre-trained embedding space. In this project, a toolkit fairseq has been used.as it offers a pre-trained embedding space.

An embedding space transfers a word to a vector, but occasionally words in a phrase have various meanings; in this case, a positional encoder comes in helpful.

Positional encoders are vectors that have information on distances between words and the sentences. Some research publications mention using the sine and cosine functions to construct this vector, however any suitable function might be used. Following the application of input embedding and positional encoding, word vectors with positional information are created. This data is processed by an encoder block, which includes a multi-head attention layer and a feed forward layer.

The attention layer is responsible for answering which part of input should be focused on. For example, while translating from English to Hindi, the question to be answered is how relevant is each word in the English sentence relevant to other words in that sentence. This is expressed in the ith attention vector and is formulated in the attention block. An attention vector can be constructed for each word, capturing contextual links between words in a phrase, which is particularly valuable in translation in the long term. The main issue is that the attention vector may be too weak, thus eight such vectors are obtained each word and a weighted average is used to generate the final attention vector for each word.

The feed-forward net is a basic feed-forward neural network that is applied to each of the attention vectors.In practise, these nets are used to transform attention vectors into a format that the following encoder or decoder block can understand.Individual attention vectors are transferred through the feed-forward net.

Each attention net is independent of each other so parallelisation can be used. Through this, all words can be passed simultaneously into the encoder block, with each word's output consisting of a sequence of vectors.

Conversely, comparable layers that perform the same functions are present in the decoder block. For instance, in the English to Hindi training phase, the Hindi sentence is supplied to the decoder as a vector rather than a word. The positional vector is added to the word once it has been processed to create its vector form, which creates the word's context inside a phrase. After that, this vector is finally sent into a decoder block, which consists of three primary components, two of which are shared with the encoder block. To show how much each word in the Hindi sentence is related to every other word in the same sentence, the self attention block creates attention vectors for each word. As one vector is obtained from each word in the Hindi and English phrases, these attention vectors are moved, together with the encoder's vectors, into another block called the encoder-decoder attention block.

This is the primary section where word mapping from Hindi to English is done. The attention vector for every word in sentences written in Hindi and English is the block's output. Every vector shows how a word in one language relates to other ones.A feed-forward unit receives each attention vector, processes it, and generates an output vector that is easier for the following linear layer or decoder block to understand. Another feed-forward connected network is the linear layer. It is employed to increase the Indic language's vocabulary. It is converted into a human-readable probability distribution by the softmax layer. The word with the greatest probability is the last one. The overall decoder does several time steps and guesses the word that will follow next in order to generate the token at the end of the sentence. In addition a sort of normalising comes after each layer.It facilitates optimization at greater learning rates.

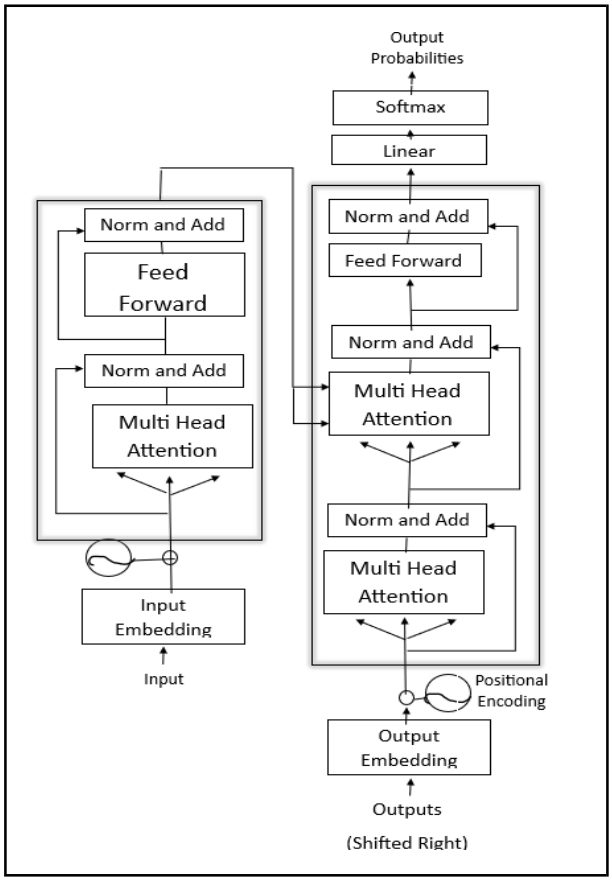

Fig 1. Transformer model

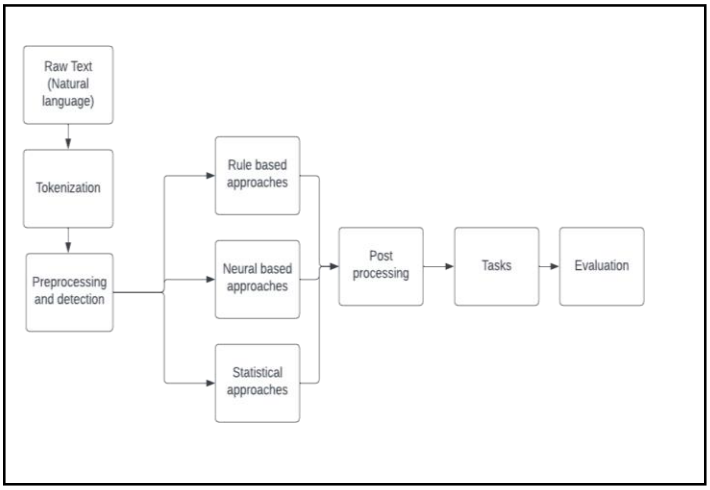

Fig 2. Design of translation process

IV. RESULTS AND DISCUSSION

The main objective was to build a model that accurately translates Indian languages to English and vice versa. In the images , it can be noticed that the right dataset list variable takes in the sentence in the source language which will be translated. A user defined function takes three parameters of dataset, source language code, target language code. For example, in Fig 3, hi_dat is the list which takes up a sentence. The variable ps_rettxt stores the result obtained from the function ansu_trans whose parameters are the list, source language being Hindi and target language being English. On printing , the English text is obtained. The same procedure follows for the other languages with their corresponding datasets and sentences.

```
#Hindi
hi_dat=['पुलिस मौके पर मौजूद है।']
ps_rettxt = ansu_trans(hi_dat,'hi','en')
print(ps_rettxt)

100%|██████████| 1/1 [00:00<00:00, 3979.42it/s]
['Police is present on the spot.']
```

Fig 3. Translation of Hindi to English

kn_dat is the list which takes up a sentence. The variable ps_rettxt stores the result obtained from the function ansu_trans whose parameters are the list, source language being Kannada and target language being English in this block. On printing , the English text is obtained.

```
#Kannada
kn_dat=['ಅರ್ಥ ಮಾಡಿಕೊಳ್ಳಬೇಕು ಅನ್ನಿಸಿದೆಯಾ?']
ps_rettxt = ansu_trans(kn_dat,'kn','en')
print(ps_rettxt)

100%|██████████| 1/1 [00:00<00:00, 3492.34it/s]
['Do you understand?']
```

Fig 4. Translation of Kannada to English

```
#Malayalam
ml_dat=['പിന്നെ ആരാണത് കഴുകുക?']
ps_rettxt = ansu_trans(ml_dat,'ml','en')
print(ps_rettxt)

100%|██████████| 1/1 [00:00<00:00, 3964.37it/s]
['Then who will do it?']
```

Fig 5. Translation of Malayalam to English

```
#Telugu
te_dat=['ఆట్టుకుళక్కు చిరిక్కాన్ నల్లరసం.']
ps_rettxt = ansu_trans(te_dat,'te','en')
print(ps_rettxt)

100%|██████████| 1/1 [00:00<00:00, 2442.81it/s]
['It is located in the center of the city.']
```

Fig 6. Translation of Telugu to English

```
#Tamil
ta_dat=['எனக்கு உதவி தேவைப்படுகிறது']
ps_rettxt = ansu_trans(ta_dat,'ta','en')
print(ps_rettxt)

100%|| 1/1 [00:00<00:00, 3927.25it/s]
['I need help']
```

Fig 7. Translation of Tamil to English

The functions and parameters are the same as in the last translation; the only change is that the source language is English, as is the dataset list variable. The target language can be specified with the 2 letter code.

```
# ENGLISH to TELAGU
en_dat =['Eat more fruits']
ps_rettxt = ansu_trans(en_dat,'en','te')
print(ps_rettxt)

100%|| 1/1 [00:00<00:00, 3116.12it/s]
['ಪಂಡ್ಲು ಎಕ್ಕುವಗಾ ತಿನಂಡಿ']
```

Fig 8. Translation of Telugu to English

```
# ENGLISH to HINDI
en_dat=['Work hard till you succeed']
ps_rettxt = ansu_trans(en_dat,'en','hi')
print(ps_rettxt)

100%|| 1/1 [00:00<00:00, 582.62it/s]
['जब तक आप सफल नहीं होते, तब तक कड़ी मेहनत करें']
```

Fig 9. Translation of Telugu to English

```
# ENGLISH to HINDI
en_dat=['Work hard till you succeed']
ps_rettxt = ansu_trans(en_dat,'en','hi')
print(ps_rettxt)

100%|| 1/1 [00:00<00:00, 582.62it/s]
['जब तक आप सफल नहीं होते, तब तक कड़ी मेहनत करें']
```

Fig 10. Translation of English to Hindi

```
# ENGLISH to KANNADA
en_dat=['Plant more trees']
ps_rettxt = ansu_trans(en_dat,'en','kn')
print(ps_rettxt)

100%|| 1/1 [00:00<00:00, 1336.62it/s]
['ಹೆಚ್ಚು ಹೆಚ್ಚು ಗಿಡಗಳನ್ನು ಬೆಳೆಸಿ']
```

Fig 11. Translation of Telugu to Kannada

Even for Indic to Indic translation we use the same model but apply corresponding codes of language in source and target parameters.

```
#Telugu to Kannada
te_dat =['ಪಂಡ್ಲು ಎಕ್ಕುವಗಾ ತಿನಂಡಿ']
ps_rettxt = ansu_trans(te_dat,'te','kn')
print(ps_rettxt)

100%|| 1/1 [00:00<00:00, 4355.46it/s]
['ಹಣ್ಣುಗಳನ್ನು ಹೆಚ್ಚು ಸೇವಿಸಿ']
```

Fig 12. Translation of Telugu to Kannada

```
#Hindi to Kannada
hi_dat =['जब तक आप सफल नहीं होते, तब तक कड़ी मेहनत करें']
ps_rettxt = ansu_trans(hi_dat,'hi','kn')
print(ps_rettxt)

100%|| 1/1 [00:00<00:00, 3581.81it/s]
['ನೀವು ಯಶಸ್ವಿಯಾಗುವವರೆಗೆ ಶ್ರಮಿಸಿ.']
```

Fig 13. Translation of Hindi to Kannada

The proposed model uses a transformer model, which helps it to get accurate translation whereas the other existing models use seq2seq models. The proposed model also uses the text to speech API which helps the user to get accurate audio output and in the local accent which will help with the communication between people, which is the primary goal.

Table 1. Comparison of the Proposed Method with Traditional Machine Translation Methods

Criteria	Proposed Method (Transformer Model using Fairseq)	Traditional Machine Translation Methods
Long-Term Dependencies	Self-attention mechanism effectively handles long-term dependency.	It may be difficult to capture long-term interdependence.
Parallel Processing	Allows for concurrent processing of input	Sequential models analyze input sequentially
Word Embeddings	Uses pre-trained embedding space	It may be necessary to train word embeddings separately.
Attention Mechanism	Multi-head attention is used to collect contextual relationships between words.	Methods, such as global attention and local attention, may be used.
Decoder Architecture	Uses a decoder block with self-attention and encoder-decoder attention layers.	Traditional models may employ RNN-based decoders or statistical machine translation techniques.
Text-to-Speech Integration	Uses a text-to-speech API for accurate audio output in the local accent.	Text-to-speech capabilities may need separate integration with traditional techniques.

The below Table 2, depicts the comparison between the BLEU metrics of our model versus few other popularly known models like OPUS , mT5 and mBart , the scores have been calculated across the WAT2021 testset to get accurate results. It is observed that our proposed model performs better than the other models that have been evaluated.

Table 2. BLEU scores for translation across WAT2021 testset comparing the proposed model with other popularly known models

Language Model	**Our Model**	**OPUS**	**mT5**	**mBART**
Kannada	36.4	-	27.8	-
Hindi	43.9	13.3	39.2	33.1
Tclugu	36.2	-	28.5	4.3
Tamil	33.2	-	27.8	26.8
Malayalam	34.6	5.7	26.8	19.1

V. CONCLUSION

Finally, this study looked at the performance of the Fairseq toolset for neural machine translation of three Indian languages: Hindi, Kannada, and Telugu. In addition, the paper investigated the efficacy of Byte Pair Encoding (BPE) and Transformer Attention based models in increasing translation quality. Fairseq, when paired with mentioned architectures, exhibited considerable gains in translation quality for all languages, according to our testing results. Furthermore, our findings show that using a large training dataset, careful hyperparameter adjustment, and adequate preprocessing approaches can improve translation quality even further. Overall, this study advances Indian language translation by providing insights into the successful application of neural machine translation systems.

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